

ADITYA PRASANNA MOGARE

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EDUCATION

- **University of Southern California** Los Angeles, CA
Master's of Science - Computer Science; **GPA: 3.70/4.0** Aug. 2024 – May 2026
- **University of Mumbai** Mumbai, India
Bachelor's of Technology -Computer Engineering; **GPA: 9.63/10** Aug. 2020 – May 2024

WORK EXPERIENCE

- **INTERDEPENDENT Studios** Los Angeles, CA
Software Engineering and AI Automation Intern May 2025 – Aug. 2025
 - **Eliminated** manual screenplay submission bottleneck by building **Flask webhook** pipeline that automates PDF ingestion, AI evaluation, and studio frame routing in Miro; processed **5000+** **creator submissions** with zero manual board ops, enabling studio reviewers to focus on quality curation rather than administrative coordination.
 - **Scaled** platform from manual review to transparent AI-driven feedback by collaborating on unified AI screenplay evaluation system (**OpenAI** + 6-category rubric); integrated **Mailgun**, **Webflow**, **Stripe Identity**, and **PostgreSQL** to automate creator feedback, script publishing, and access tier management—enabling creators to understand decisions and reducing studio overhead.
 - **Deployed** Flask backend (Gunicorn) serving Miro virtual studio platform; implemented creator identity modeling (Stripe verification, vote budgets, access tiers) supporting gated submissions and community voting features.
- **University of Southern California** Los Angeles, CA
Graduate Research Intern Jan. 2025 – Jun. 2025
 - **Designed** PyTorch CNNs (**ResNet50**) for **OCTA**-based diabetic retinopathy detection, utilizing **CLAHE preprocessing** and real-time augmentation to achieve **85%** validation accuracy.
 - **Enhanced** rare disease classification by mitigating dataset class imbalance, leveraging dynamic class weighting and strategic downsampling to boost minority-class recall by **30%**.
 - **Containerized** training/inference pipelines on **AWS EC2 (S3 storage)**, improving **ROC-AUC by 5%** and reducing training time through efficient batch scheduling.
- **HiringTek** Hyderabad (Remote)
Software Engineering and AI Intern Aug. 2022 – Aug 2023
 - **Engineered** GPT-3 question-generation microservice with **sub-20ms** response times via **Flask** + **MongoDB caching**, enabling instant AI-powered candidate interviews at scale.
 - **Reduced** data validation errors by **25%** through **MongoDB pipeline** optimization and backend refactoring, improving system reliability and scalability.
 - **Built** 4 core frontend components in **React** + **TypeScript** with Jest E2E testing, ensuring cross-module reliability and improved admin usability.

PROJECTS

- **AgentGuard - Open-source multi-agent AI observability platform** **Github Link**
Python SDK, PostgreSQL, Django, Redis , Splunk MCP server May 2026 – Present
 - Identified **observability gap** in multi-agent AI systems; engineered AgentGuard—a production-grade Python SDK + Django backend that instruments agent execution with zero-overhead tracing decorators (@trace_agent, @trace_tool), **streams telemetry to Splunk HEC**, and surfaces failures via real-time dashboards.
 - Built dual export architecture (Splunk HEC + Django REST API) supporting enterprise and backend-only deployments; containerized full stack via Docker Compose with **PostgreSQL**, **Redis**, and optional Splunk integration—enabling <5-minute production deployment for **multi-agent teams**.
- **Lucerna - Defense Intelligence Platform (DoD)** **Github Link**
React, Django, PostgreSQL, Redis, Celery, Docker, LLMs, Text-to-SQL Jan 2026- May 2026
 - **Built Data Intelligence Platform** for U.S. Air Force (**USC H4D program**), eliminating procurement visibility gap across legacy ERP systems via Text-to-SQL engine.
 - **Translated** natural language analytics queries into optimized SQL via a **Text-to-SQL intelligence engine**, executing against an OLAP database and passing results through a hosted LLM to auto-flag high-risk and expiring U.S. Air Force contracts

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL, C,C++, Java
- **Backend & APIs:** Flask, Django, FastAPI, Node.js/Express, RESTful APIs, Webhooks, Stripe & Third-party API integrations, JWT/OAuth, Async patterns, WSGI (Gunicorn)
- **Databases & Caching:** PostgreSQL (SQLAlchemy, Alembic), MongoDB, Redis (pub/sub, caching), DynamoDB, Query optimization, Database design.
- **Cloud & DevOps:** AWS (EC2, S3, RDS), Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions), Microservices architecture, Event-driven systems, Celery (task queues)
- **AI/ML & LLMs:** PyTorch, Hugging Face Transformers, LoRA, RAG (FAISS, BM25), LLaMA, Mistral, Prompt engineering, MCP (Model Context Protocol), OpenAI APIs, vLLM